CO2 EXPERIMENT 09

Simulation of Telemetry Data Ingestion to Time-Series Database and Visualization using Grafana, Power BI and Tableau

AIM

To simulate the ingestion of telemetry data from IoE sensors into a time-series database and visualize the collected data using Grafana, Power BI and Tableau for monitoring and performance analysis.

2. OBJECTIVES

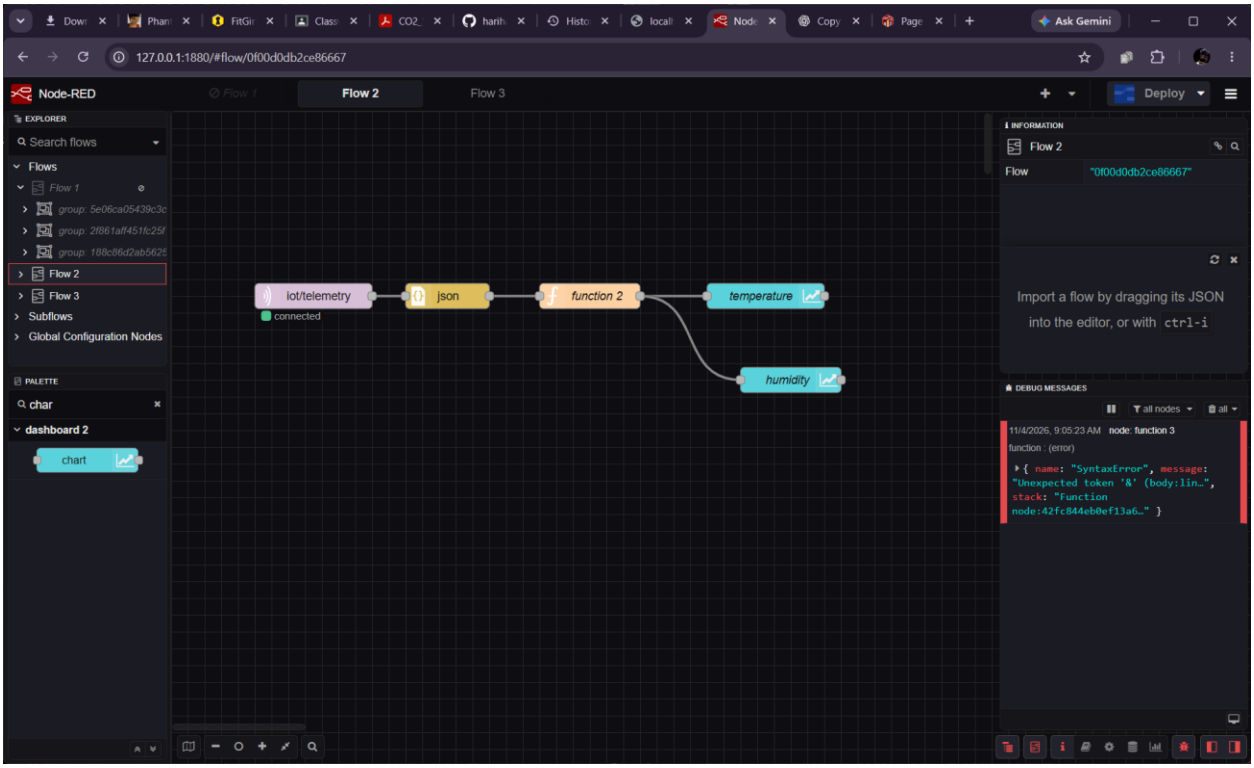
1. To understand telemetry data collection in Internet of Everything (IoE) applications.
2. To simulate real-time data generation from multiple sensors.
3. To study the process of ingesting sensor data into a time-series database.
4. To analyse time-based sensor readings such as temperature, humidity and pressure.
5. To create dashboards using Grafana, Power BI and Tableau.
6. To compare the visualization capabilities of different dashboard platforms.
7. To identify trends, variations and abnormal conditions in sensor data.

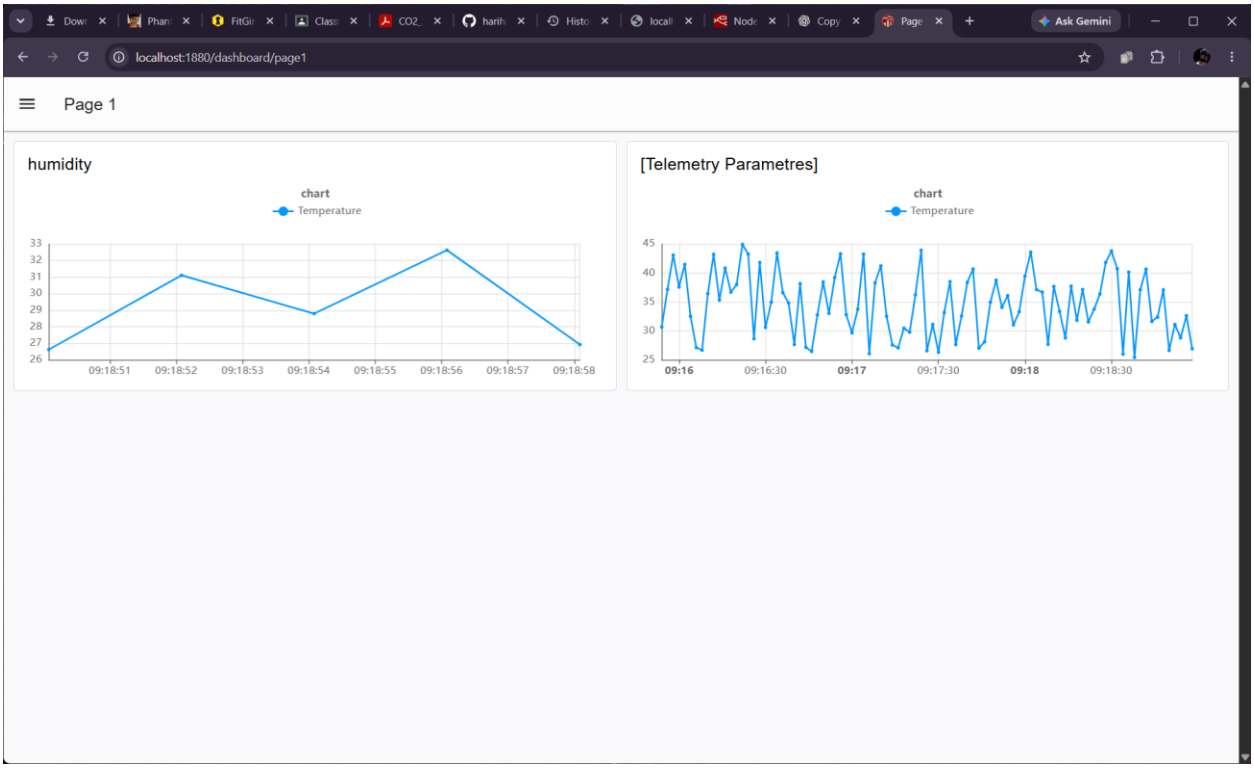
3. PROCEDURE

1. Open Python IDLE or Google Colab and import the required libraries such as NumPy, Pandas and Matplotlib.
2. Define the simulation parameters, including the number of IoE sensors and data collection intervals.
3. Generate simulated telemetry data such as temperature, humidity and pressure with timestamps.
4. Store the generated sensor readings in a Pandas DataFrame.
5. Prepare the telemetry data in a structured format for database ingestion.

6. Connect the telemetry data to a time-series database such as InfluxDB.
7. Insert the sensor readings into the database with timestamps and sensor identifiers.
8. Retrieve the stored data and perform basic analysis to identify trends and variations.
9. Connect Grafana to the time-series database and create real-time monitoring dashboards.
10. Import the collected data into Power BI and Tableau to create charts, graphs and analytical dashboards.
11. Compare the visualization features, monitoring capabilities and analytical performance of all three platforms.

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4. RESULT

The simulation of telemetry data ingestion into a time-series database and visualization using Grafana, Power BI and Tableau was successfully studied and performed.

The simulated sensor data was generated, organized and prepared for database storage. The visualization platforms were used to represent telemetry readings through charts, graphs and dashboards.

Grafana is suitable for real-time monitoring, while Power BI and Tableau provide interactive data analysis and reporting capabilities